GenAI. Lecture 3 - WGAN, Optimal Transport problem

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Recap

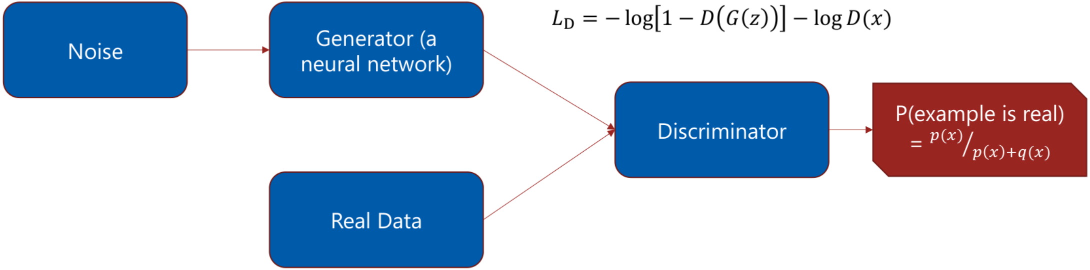
Recap. Key points

- in practice, $\sup_T$ is replaced with "torch" optimization (maximization) of variational parameters η of variational neural network T_η
- on practice, D can be trained by BCE in most cases ...
- ... moreover, if D estimates $P(x)$, then BCE minimization identical to $KL(Q|P)$ minimization ...
- ... if D estimates $p(x \sim P|x)$ $(p(real|x))$, then BCE minimization identical to $JS(P|Q)$ minimization
- GAN is unstable, but really fast in practice
- and many many GAN tricks

Recap. GAN

$$L_{\mathrm{G}} = -\log D(G(z))$$

$$L_{\mathrm{D}} = -\log[1 - D(G(z))] - \log D(x)$$



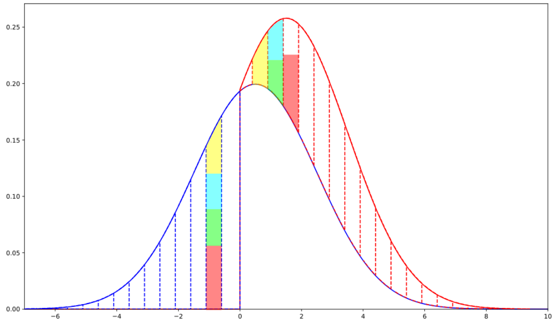
Wasserstein distance

Wasserstein distance

- Distributions $P(x)$ and $Q(x)$ are viewed as describing the **amounts of "dirt" at point** x.
- We want to convert one distribution into the other by **moving around some amounts of dirt**.
- The cost of moving an amount m from x_1 to x_2 is:

$$m \cdot \|x_2 - x_1\|.$$

- EMD$(P, Q) =$
  minimum total cost of converting P into Q.



Wasserstein distance

- Say, we have a moving plan $\gamma(x_1, x_2) \geq 0$:

$$\gamma(x_1, x_2)\, dx_1\, dx_2 \quad \text{– how much dirt we're moving from } [x_1, x_1 + dx_1] \text{ to } [x_2, x_2 + dx_2].$$

- Then, the cost of moving from $[x_1, x_1 + dx_1]$ to $[x_2, x_2 + dx_2]$ is:

$$\|x_2 - x_1\| \cdot \gamma(x_1, x_2)\, dx_1\, dx_2.$$

- And the total cost is:

$$C = \int_{x_1, x_2} \|x_2 - x_1\| \cdot \gamma(x_1, x_2)\, dx_1\, dx_2 = \mathbb{E}_{x_1, x_2 \sim \gamma(x_1, x_2)} \|x_2 - x_1\|.$$

- Since we want to convert P to Q, the plan has to satisfy: $\int_{x_1} \gamma(x_1, x_2)\, dx_1 = Q(x_2), \quad \int_{x_2} \gamma(x_1, x_2)\, dx_2 = P(x_1).$

Wasserstein distance

- Let π be the set of all plans that convert P to Q, i.e.:

$$\pi = \left\{ \gamma : \gamma \geq 0, \quad \int_{x_1} \gamma(x_1, x_2) dx_1 = Q(x_2), \quad \int_{x_2} \gamma(x_1, x_2) dx_2 = P(x_1) \right\}.$$

- Then, the Wasserstein distance between P and Q is:

$$\mathrm{EMD}(P, Q) = \inf_{\gamma \in \pi} \mathbb{E}_{x_1, x_2 \sim \gamma} \|x_2 - x_1\|.$$

- **Optimization over all transport plans is not too friendly**

Wasserstein distance

For the continuous case, there are a set of **p-Wasserstein distances**, with $W_p(p_x, q_y)$ defined for $x \in M$, $y \in M$, and a distance D on x, y:

$$W_p(p_x, q_y) = \inf_{\gamma \in \Pi(x,y)} \int_{M \times M} D(x, y)^p \, d\gamma(x, y),$$

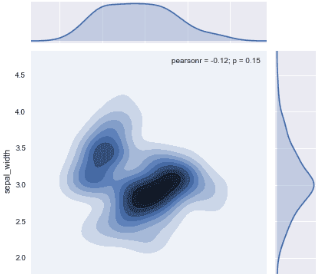
where $\Pi(x, y)$ is the set of all joint distributions having p_x and q_y as their marginals.

Wasserstein distance

In particular, W_1 distance with Euclidean norm is:

$$W(p_x, q_y) = \inf_{\gamma \in \Pi(x,y)} \int_{M \times M} D(x, y)\, d\gamma(x, y) = \inf_{\gamma \in \Pi(x,y)} \mathbb{E}(\|x - y\|),$$

Which brings an evident connection to EMD.



Explanation:

- Two-dimensional representation of the transport plan between horizontal (μ) and vertical (ν) PDFs.
- This is **not a unique plan**.
- The infimum must be taken

Wasserstein distance. Lipschitz Continuity

- f is **Lipschitz-k continuous** if:
- There exists a constant $k \geq 0$ such that for all x_1 and x_2:

$$|f(x_1) - f(x_2)| \leq k \cdot \|x_1 - x_2\|.$$

- This inequality ensures that the function f does not change too rapidly, bounded by k.

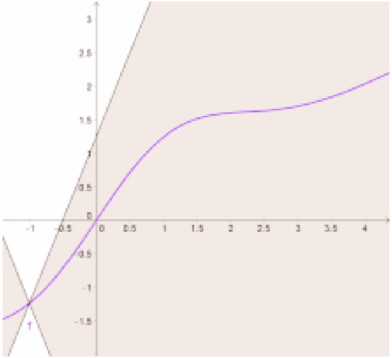


Image from https://en.wikipedia.org/wiki/Lipschitz_continuity

Wasserstein distance. Kantorovich-Rubinstein Duality

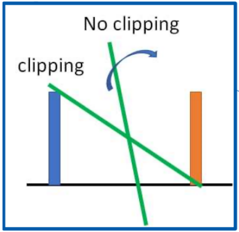
- P: True PDF, Q: Fitted PDF.

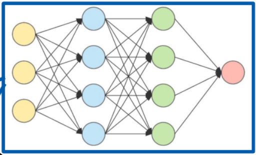
$$W(P;Q) = \sup_{f \in Lip_1} \left(\mathbb{E}_{x \sim P}[f(x)] - \mathbb{E}_{x \sim Q}[f(x)] \right),$$

where Lip_1 is the set of 1-Lipschitz functions.

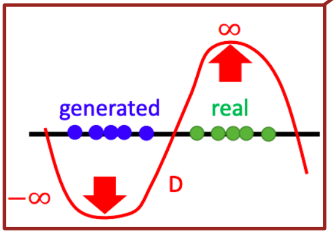
WGAN

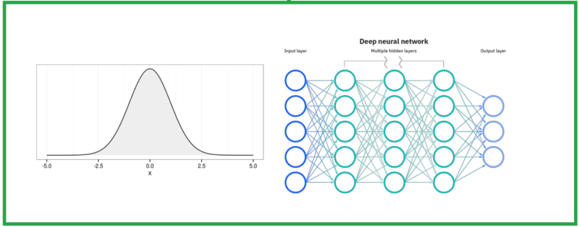
WGAN





$$W(P;Q) = \sup_{f \in Lip_1} \left(\mathbb{E}_{x \sim P} f(x) - \mathbb{E}_{x \sim Q} f(x) \right),$$





Lipschitz-1 Condition and Neural Networks

$$W(P;Q) = \sup_{f \in Lip_1} \left(\mathbb{E}_{x\sim P}[f(x)] - \mathbb{E}_{x\sim Q}[f(x)] \right),$$

- f is a neural net – **discriminator** (*'critic'* in the original paper).
- The expectations are estimated from samples.
- Lipschitz-1 continuity can be replaced with Lipschitz-k continuity:
  - Estimate $k \times W(P,Q)$.
  - Achieved by **clipping the weights** of the critic:

$$w \to \text{clip}(w, -c, c),$$

with some constant c.

WGAN

Algorithm 1: WGAN, our proposed algorithm. All experiments in the paper used the default values: $\alpha = 0.00005$, $c = 0.01$, $m = 64$, $n_{\text{critic}} = 5$.

Require: α: learning rate, c: clipping parameter, m: batch size, n_{critic}: number of iterations of the critic per generator iteration.

Require: w_0: initial critic parameters, θ_0: initial generator parameters.

while θ has not converged **do**

  for $t = 0, \ldots, n_{\text{critic}}$ **do**

    Sample $\{x^{(i)}\}_{i=1}^{m} \sim \mathbb{P}_r$: a batch from the real data.

    Sample $\{z^{(i)}\}_{i=1}^{m} \sim p(z)$: a batch of prior samples.

    $g_w \leftarrow \nabla_w \left[\frac{1}{m}\sum_{i=1}^{m} f_w(x^{(i)}) - \frac{1}{m}\sum_{i=1}^{m} f_w(g_\theta(z^{(i)}))\right]$

    $w \leftarrow w + \alpha \cdot \text{RMSProp}(w, g_w)$

    $w \leftarrow \text{clip}(w, -c, c)$

  end for

  Sample $\{z^{(i)}\}_{i=1}^{m} \sim p(z)$: a batch of prior samples.

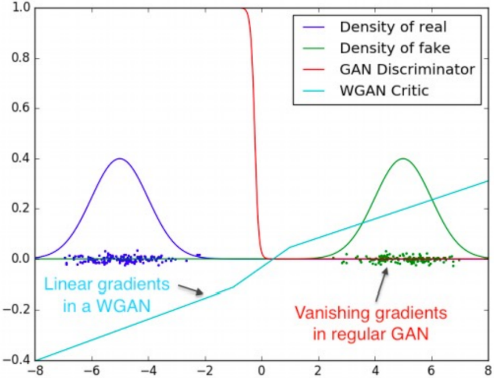
  $g_\theta \leftarrow -\nabla_\theta \frac{1}{m}\sum_{i=1}^{m} f_w(g_\theta(z^{(i)}))$

  $\theta \leftarrow \theta - \alpha \cdot \text{RMSProp}(\theta, g_\theta)$

end while

WGAN: problems solved

- The **vanishing gradient problem** is solved.
- The **mode collapse problem** is addressed.
- From authors: *"Weight clipping is a clearly terrible way to enforce a Lipschitz constraint."*

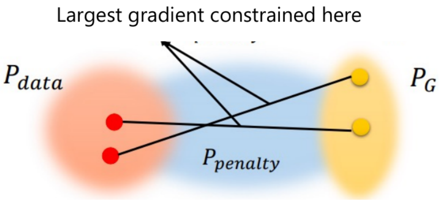


WGAN-GP

- **Weight clipping** makes the critic less expressive and the training harder to converge.
- Optimal f should satisfy $\|\nabla f\| = 1$ almost everywhere under P and Q.
- Also: $\|f\|_L \leq 1 \iff \|\nabla f\| \leq 1$.
- Can replace weight clipping with a gradient penalty term:

$$\mathrm{GP} = \lambda \int \max[(\|\nabla_{\tilde{x}} f(\tilde{x})\| - 1)^2] \, dx$$

$$\mathrm{GP} = \lambda \mathbb{E}_{\tilde{x} \sim \mathbb{P}_{\tilde{x}}}[(\|\nabla_{\tilde{x}} f(\tilde{x})\| - 1)^2].$$



$$\mathbb{P}_{\tilde{x}} : \begin{cases} \tilde{x} = \alpha x_1 + (1-\alpha) x_2, \\ \alpha \sim \mathrm{Uniform}(0,1), \\ x_1 \sim P, \quad x_2 \sim Q. \end{cases}$$

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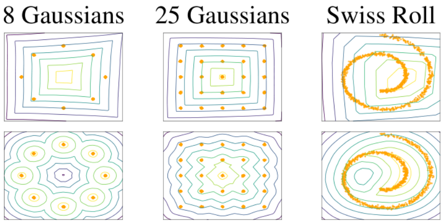

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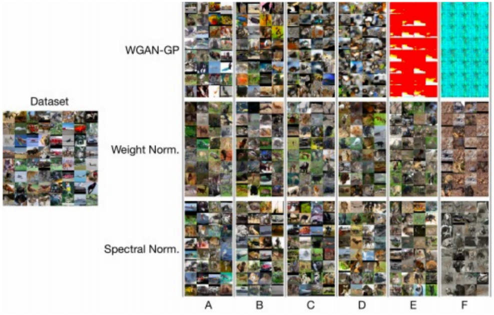
WGAN-SN

- Spectral normalization proposes to use normalized weights:
$$W_{SN} = \frac{W}{\sigma(W)}$$
where:
$$\sigma(W) = \max_{h:h\neq 0} \frac{\|Wh\|_2}{\|h\|_2}.$$
- This gives constraints on the gradient:
$$\|f\|_{\text{Lip}} \leq \prod_{i=1}^{l} \sigma(W_l).$$



(b) STL-10

Figure 6: Generated images on different methods: WGAN-GP, weight normalization, and spectral normalization on CIFAR-10 and STL-10.

Miyato et al. Spectral Normalization for Generative Adversarial Networks, ICLR 2018

Conclusion

- **WGAN is a powerful generative model.**
- Simpler training procedure but requires controlling Lipschitz continuity.
- Several ideas exist to achieve this.
- **Still problems:**
  - Kantorovich-Rubinstein duality is only mimicked.
  - Gradient gets stuck near solutions.